Transportation Safety and Environment, 2024, **6**: tdad026
DOI: 10.1093/tse/tdad026
Advance access publication date: 18 May 2023
Research Article

Ride-hailing origin-destination demand prediction with spatiotemporal information fusion

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Abstract

Accurate demand forecasting for online ride-hailing contributes to balancing traffic supply and demand, and improving the service level of ride-hailing platforms. In contrast to previous studies, which have primarily focused on the inflow or outflow demands of each zone, this study proposes a conditional generative adversarial network with a Wasserstein divergence objective (CWGAN-div) to predict ride-hailing origin-destination (OD) demand matrices. Residual blocks and refined loss functions help to enhance the stability of model training. Interpretable conditional information is employed to capture external spatiotemporal dependencies and guide the model towards generating more precise results. Empirical analysis using ride-hailing data from Manhattan, New York City, demonstrates that our proposed CWGAN-div model can effectively predict the network-wide OD matrix and exhibits strong convergence performance. Comparative experiments also show that the CWGAN-div outperforms other benchmarking methods. Consequently, the proposed model displays potential for network-wide ride-hailing OD demand prediction.

Keywords: intelligent transport system; ride-hailing; generative adversarial networks; spatiotemporal dependencies; origin-destination (OD) demand prediction

1. Introduction

The development of mobile internet and GPS technology has given rise to on-demand ride-hailing as a predominant mode of urban transportation due to its convenience and flexibility. Ride-hailing services have transformed the entire landscape of the taxi industry and improved the balance between supply and demand [1]. Nonetheless, the current efficiency of ride-hailing services offers significant scope for improvement [2]. GPS trajectories obtained from ride-hailing transactions (e.g., Didi Chuxing, Uber) enable the analysis of zone-based and origin-destination (OD)-based demand [3]. Zone-based demand indicates destination-unknown passenger demand originating in each traffic zone, while OD-based demand represents the passenger demand from the origin zone to the destination zone, illustrating the interconnectivity between zones and emphasizing specific demand distribution. Accurate OD demand prediction can assist traffic managers to estimate traffic conditions and devise traffic control measures. For operators, the results of OD prediction can help with vacant vehicle re-allocation and increase platforms' profits [4].

Most existing studies only predict the zone-based demand and do not have the capability of matching origins with destinations [5–10]. The OD demand is characterized by high data volume, high dimensionality and high sparsity [11]. Specifically, the number of network-wide OD pairs grows rapidly due to the vast amounts of online ride-hailing orders. Meanwhile, demand varies considerably among these OD pairs, attributable to urban centres' agglomeration effects and the limited number of long-distance trips, resulting in a highly sparse OD matrix. Moreover, OD demand fluctuates in different time periods (peak time, off-peak time) and days (weekdays, holidays). Consequently, forecasting network-wide OD demand poses a greater challenge than predicting zone-based demand. The primary motivation for this paper is to achieve a highly accurate network-wide OD demand prediction by mining internal spatiotemporal correlations and external spatiotemporal dependencies.

In the past, studies on OD demand prediction primarily relied on static models, which projected fixed OD demand based on previous research and investigations [12, 13]. However, static OD prediction is unable to provide timely traffic data and often yields poor results due to the complexity and variability of modern transportation systems. To tackle this issue, Toledo and Kolechkina [14] proposed a general solution scheme that used linear approximations of the assignment matrix and mapped OD demand to link traffic counts for dynamic OD estimation. Barceló et al. [15]. applied the Kalman filter approach to estimate OD matrices in corridors. However, these methods neglect the impact of external spatiotemporal dependencies on OD demand. They are less effective when the temporal characteristics of OD demand become insignificant or when traffic conditions undergo abrupt changes.

In recent years, machine learning methods have become increasingly popular among researchers in the transportation field [16–18]. For instance, Zhang and Zhao [16]. proposed a clustering ensemble framework that divided OD pairs into different clusters and used machine learning models to build the cluster-specific models for OD demand prediction. However, due to the model's complexity, the study did not incorporate temporal features as variables for predicting OD demand, and the improvement in

Received: April 23, 2023. **Revised:** May 12, 2023. **Accepted:** May 13, 2023

model prediction accuracy was limited when the spatial heterogeneity over different zones was high. Golshanrad et al. [17]. used four-dimensional *K*-means clustering combined with a stacked recurrent neural network to avoid the information loss caused by two-dimensional clustering, and demonstrated the importance of incorporating external factors such as temporal characteristics. In terms of prediction accuracy, these machine learning methods attain some improvement over traditional mathematical models. However, in practice, they often necessitate converting the OD matrix into a one-dimensional vector input model, which may not always be suitable for high-dimensional and sparse urban road network traffic data.

As an important branch of machine learning approaches, deep learning has attracted considerable attention from the academic community due to its remarkable capacity to extract abstract features from large datasets. Deep learning-based techniques have been widely applied in traffic prediction research [4, 11, 19–23]. Zhang et al. [11]. proposed a channel-wise attentive split–convolutional neural network (CAS-CNN) model to predict short-term OD demand in urban rail transit; however, this method only mined patterns of historical demand data without considering spatial features affecting OD demand, such as network topology and inter-station distance. Wang et al. [4]. partitioned the study area into regular grids to define different traffic zones, but this approach may separate regions exhibiting spatiotemporal homogeneity in traffic demand. Moreover, the long short-term memory network (LSTM) employed is better suited for low-dimensional, regularly structured sequence data, and fails to thoroughly explore spatiotemporal features in high-dimensional OD demand. To capture both temporal and spatial information in traffic data, Narmadha and Vijayakumar [19] used a combined LSTM and CNN model for short-term traffic flow prediction. Although these deep learning methods can access the internal spatiotemporal correlations of OD demand, they do not take into account external spatiotemporal dependencies and offer limited interpretability.

Generative adversarial networks (GANs) [24] represent a state-of-the-art deep learning technique specifically tailored for optimizing generative tasks. The basic idea of the initial GAN is to generate samples that cannot be distinguished from real data through adversarial training of the generator and discriminator. Following the debut of the initial GAN, numerous versions have been proposed, offering enhancements to the network architecture. For example, Mirza and Osindero [25]. proposed the conditional GAN (CGAN), which learns to generate samples by incorporating specific conditional information or given features. Radford et al. [26]. integrated convolutional neural networks (CNNs) into the GAN's structure and proposed the deep convolutional generative adversarial network (DCGAN), which has become a benchmark for developing subsequent GAN models and is widely employed in image processing. The WGAN [27] altered the initial GAN's objective function to incorporate a Wasserstein metric, significantly enhancing the initial GAN's training stability. However, the WGAN necessitates meeting the Lipschitz constraint, which poses challenges in practical applications. To address this issue, Wu et al. [28]. introduced a Wasserstein divergence objective for GANs (WGAN-div) and demonstrated through mathematical derivation that the WGAN-div can achieve the WGAN's high performance without having to satisfy the Lipschitz constraint.

Due to their potent generative capabilities, GANs have been introduced into the transportation field by some researchers for traffic prediction and estimation [29–35]. Zhang et al. [29] first employed GAN to estimate trip travel time distributions in an urban road network. Zhang et al. [30] proposed the travel times imputation GAN (TTI-GAN), which can generate travel times for links without sufficient observations by modelling travel time distributions for links with rich data. Yu et al. [31] proposed a deep learning framework combining a modified DBSCAN model and a CGAN model to predict the taxi-passenger demand that considers spatiotemporal correlations and external dependencies. Jin et al. [32] used a WGAN for robust data-driven traffic modelling to achieve road network traffic prediction considering spatiotemporal correlations. These successful cases illustrate that GANs can effectively handle large volumes of traffic data and offer a flexible structure that can be combined with various deep learning methods to fully exploit spatiotemporal correlations in traffic data. Furthermore, by incorporating external spatiotemporal information, the prediction accuracy of the model can be further enhanced.

Inspired by the methods mentioned above, this paper processes OD demand data into spatiotemporal images and proposes a CGAN with a Wasserstein divergence objective (CWGAN-div) for ride-sourcing OD demand prediction. Unlike previous deep learning models such as CNN, LSTM and GAN, the CWGAN-div with advanced structures can further capture the spatiotemporal correlations of OD demand during the training process, and the introduced conditional data is used to represent external spatiotemporal dependencies and guides the model to gradually generate high-resolution spatiotemporal images.

The main contributions of this study include three aspects: (1) The proposed CWGAN-div merges the benefits of both CGAN and WGAN-div to ensure stable model training while considering external spatiotemporal information, enabling accurate prediction of network-wide OD demand. (2) The Moran's *I* correlation matrix is devised to capture spatial dependencies, and the *Timestamp∗Day* variable is introduced to describe temporal dependencies. (3) The residual network-based model structure aids in accelerating model convergence and reducing the risk of network degradation.

The remainder of this paper is organized as follows. Section 2 introduces preliminary information and delineates the fundamental concepts of the OD demand prediction task, while also reviewing some classic GANs. Section 3 describes the methodology and framework of the CWGAN-div. Section 4 validates the performance of the CWGAN-div through numerical experiments. Finally, Section 5 summarizes and concludes this study.

2. Preliminaries

2.1. Problem definition

The research problem addressed in this paper concerns short-term ride-hailing OD demand prediction, incorporating spatiotemporal information fusion. Divide the study site into Z traffic zones based on the urban road network geometry, and hence the size of the OD demand matrix is $Z \times Z$. Let $\boldsymbol{M}_t = \{m_t^{i,j} \mid i = 1, 2, \ldots, Z; j = 1, 2, \ldots, Z\} \in \mathbf{N}^{Z\times Z}$ represent the observed OD matrix during time interval t, where $m_t^{i,j}$ signifies the ride-hailing demand from zone i to zone j within interval t. The study's objective is to predict the OD demand matrix $\boldsymbol{M}_{t+1}$ for the subsequent time interval $t+1$, utilizing historical observations $\boldsymbol{M}_t^G = [\boldsymbol{M}_{t-T+1}, \boldsymbol{M}_{t-T+2}, \ldots, \boldsymbol{M}_t]$ from the past T time intervals and integrating temporal and spatial dependencies. The predicted value is denoted by $\hat{\boldsymbol{M}}_{t+1}$.

2.2. Spatiotemporal dependencies

Influenced by people's activity patterns, traffic demand data exhibits specific regularities in time and space [36–38]. Generally, two types of temporal correlations exist in OD demand: short-

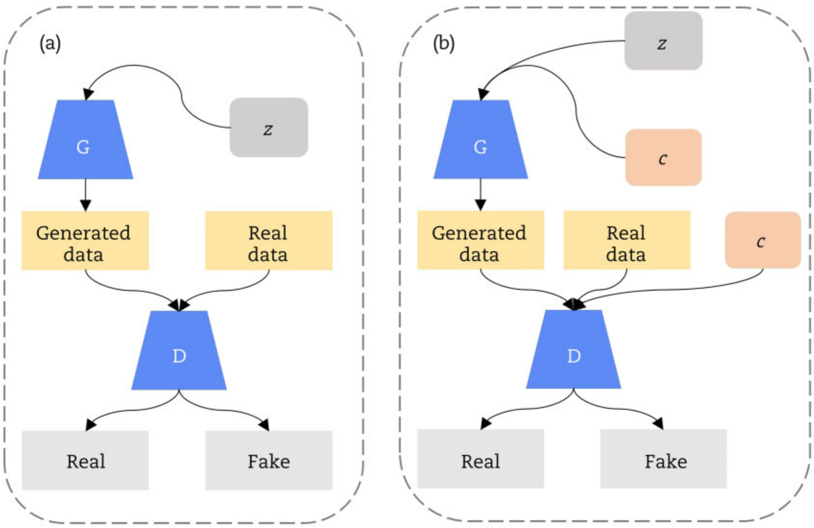

Fig. 1. General structures of (a) initial GAN and (b) CGAN.

term trend correlation and long-term period correlation [9]. The former implies that passenger demand is affected by historical demand, while the latter highlights that the demand is comparable to that in the same interval on the previous day and in the prior week. To enhance the prediction accuracy of the model, this paper considers both temporal correlations. For short-term trend correlation, historical OD demand serves as model input; for long-term period correlation, the *Timestamp∗Day* variable, proposed by Zhang et al. [29], is employed as temporal conditional information. This variable utilizes one-hot encoding to record timestamp and day information for each time interval.

Furthermore, two types of spatial correlations exist among traffic zones [39]. The first is the geographical relationship, i.e., the interaction between neighbouring zones. Generally, the closer two zones are geographically, the stronger the correlation of traffic demand between them. The distance matrix is a classic tool used to measure the correlation between regions, which is formulated as:

$$[\mathbf{A}_D]_{i,j} = \frac{1}{\mathrm{Dist}\left(lng_i, lat_i, lng_j, lat_j\right)} \tag{1}$$

where lng_i, lat_i, lng_j and lat_j represent the longitudes and latitudes of the central points of zone i and j respectively, $[\mathbf{A}_D]_{i,j}$ denotes the element of distance matrix $\mathbf{A}_D$ in the ith row and jth column and Dist calculates the distance between point (lng_i, lat_i) and (lng_j, lat_j).

Besides geographical relationships, traffic demand among different zones also exhibits a semantic relationship, such as land-use characteristics and historical demand trends. However, given the dynamic nature of traffic demand, a fixed geographical information does not consistently capture the spatial correlations among zones for each time interval, and non-geographical semantic information is often challenging to access and utilize.

Drawing on these two spatial correlations and taking inspiration from disciplines such as geography and economics, this paper proposes a novel measure of spatial dependencies that integrates both demand data and geographical information, employing the Moran's I [40, 41] as a key component of spatial dependencies calculation. The Local Moran's I is a measure that quantifies the specific characteristics of local spatial aggregation or differentiation. The Local Moran's I_i for zone i is computed as follows:

$$I_i = \frac{z_i}{S_i^2} \times \sum_{j=1, j\neq i}^{n} w_{i,j} z_j \tag{2}$$

$$z_i = x_i - \bar{x} \tag{3}$$

$$S_i^2 = \frac{\sum_{j=1, j\neq i}^{n} z_j^2}{n-1} \tag{4}$$

where n is the total number of zones, $w_{i,j}$ represents the weight of the distance matrix between zone i and zone j, x_i represents the ride-hailing demand in zone i and $\bar{x}$ is the mean ride-hailing demand value across all zones. If $I_i > 0$, this indicates a positive correlation between zone i and its surrounding zones. A higher value of I_i implies a stronger local positive correlation. An I_i value of zero denotes no correlation between zone i and its surroundings. Conversely, if $I_i < 0$, this implies a negative correlation between zone i and its surrounding zones. A smaller value of I_i indicates a stronger local negative correlation.

The Moran's I integrates both geographical relationships and the intrinsic properties of the demand data. To better represent the correlations between zones, a Moran's I correlation matrix is constructed. Specifically, the correlation between a single zone and its surrounding zones is initially calculated using the Local Moran's I, which is then employed as a coefficient and multiplied with the weights of the distance matrix, as calculated below:

$$[\mathbf{A}_M^t]_{i,j} = I_i \times [\mathbf{A}_D]_{i,j} \tag{5}$$

As evident from Equation (5), the Moran's I correlation matrix encompasses not only the fixed spatial dependencies among zones, preserving some geographical information, but also the demand correlation between the zones and their adjacent zones.

2.3. GANs

As a generative model, GAN is mainly used to generate samples from complicated probability distributions. As shown in Fig. 1(a), initial GAN consists of two parts: a generator G and a discriminator D. The generator G learns from the input data and tries to generate data indistinguishable from real data. The discriminator D is used to determine the probability that the input data comes from real data rather than generated data. The adversarial training of G and D follows the idea of a two-player zero-sum game

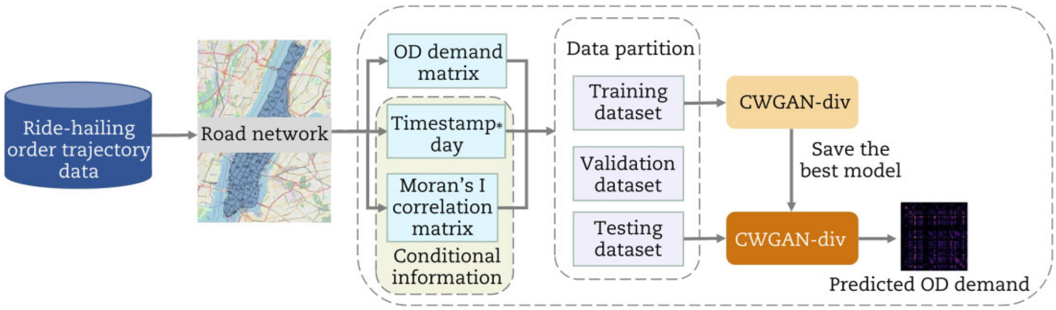

Fig. 2. Methodology framework.

with the objective function as:

$$\min_G \max_D V(G, D) = E_{x\sim P_{real}}[\log(D(x))] + E_{z\sim P_z}[\log(1 - D(G(z)))] \tag{6}$$

where x is a sample from the real data distribution P_{real} and z is a noise sample from the distribution P_z (e.g., a uniform or Gaussian distribution).

Despite the remarkable generative capabilities of the initial GAN, challenges such as training difficulties and vanishing gradients may arise in practical applications. Presently, numerous GAN variants exist to address distinct issues, and selecting an appropriate GAN along with designing a rational network structure is vital for the model stability and prediction results. CGAN, an extension of the initial GAN, incorporates external information into the model, which is beneficial for enhancing the generated results [25]. In this paper, this supplementary information can be external spatiotemporal data upon which the OD demands rely. As illustrated in Fig. 1(b), CGAN introduces conditional information c to both the generator and the discriminator compared to initial GAN. The objective function of CGAN can be expressed as:

$$\min_G \max_D V(G, D) = E_{x\sim P_{real}}[\log(D(x|c))] + E_{z\sim P_z}[\log(1 - D(G(z|c)|c))] \tag{7}$$

The WGAN [27] provides a more stable GAN framework that eliminates the need to carefully balance the training extent of G and D. The core idea of WGAN involves modifying the objective function from a Kullback-Leibler (KL) divergence to the Wasserstein metric in order to gauge the discrepancy between the generated data distribution P_g and the real data distribution P_{real}. Wasserstein distance can provide a meaningful gradient in high-dimensional space if two distributions do not overlap, or the overlap is negligible. Meanwhile, Wasserstein distance also provides a metric for GAN training, which can be expressed as:

$$W(P_{real}, P_g) = \frac{1}{K} \max_{w:\|f_w\|_L \leqslant K} E_{x\sim P_{real}}[f_w(x)] - E_{x\sim P_g}[f_w(x)] \tag{8}$$

where K is a Lipschitz constant of the function f, w represents a parameter of the function f and $\|f_w\|_L \leqslant K$ denotes a K-Lipschitz constraint required by the Wasserstein metric.

However, it is not easy to strictly impose the K-Lipschitz constraint on GAN, so most methods are often unsatisfactory, such as weight clipping and gradient penalty. Fortunately, Wu et al. [28] presented a Wasserstein divergence objective for GANs (WGAN-div), which eliminates the K-Lipschitz constraint while preserving the desirable properties of the Wasserstein distance. By incorporating Wasserstein divergence into the GAN framework, the objective function is formulated as:

$$\min_G \max_D V(G, D) = E_{x\sim P_{real}}[D(x)] - E_{G(z)\sim P_g}[D(G(z))] - kE_{x\sim P_u}[\|\nabla_x D(x)\|^p] \tag{9}$$

where k and p are parameters fine-tuned by researchers, and P_u represents a relaxed probability distribution. P_u can consist of different sampling strategies derived from P_g and P_{real} together.

Based on the preceding analysis, this study considers external spatiotemporal dependencies as the conditional information input, utilizes Wasserstein distance to measure the difference between the real data distribution and the generated data distribution and integrates the W-div into the discriminator loss function within the proposed CWGAN-div.

3. Methodology

We now introduce the CWGAN-div for predicting ride-hailing OD demand, which combines CGAN [25] and WGAN-div [28]. The methodology framework is presented in Fig. 2. First, the ride-hailing order trajectory data is mapped to the corresponding road network and traffic zones. Second, the OD demand between zones, the *Timestamp∗Day* variables and the Moran's I correlation matrix are calculated for each time interval. Third, the experimental dataset is partitioned into training, validation and test datasets; the training dataset is employed to train the CWGAN-div model, while the validation dataset is utilized to refine the model parameters. Finally, the model is trained, and the test dataset is input into the optimal model to generate the final prediction results.

3.1. CWGAN-div

This subsection develops the CWGAN-div model for ride-hailing OD demand prediction. Fig. 3 illustrates the internal structure of CWGAN-div. A concise description of the data flow is as follows. The historical OD demand spanning the preceding T time intervals is represented by $\boldsymbol{M}_t^G$, having dimensions (T, Z, Z). The temporal conditional information for each time interval is represented by $\boldsymbol{L}_t$ with a size of (H_L), that is, the length of one-hot encoding for the *Timestamp∗Day* variable. The spatial conditional information is denoted as $\boldsymbol{A}_M^t$ with a size of (T, Z, Z). $\boldsymbol{M}_t^G$, $\boldsymbol{L}_t$ and $\boldsymbol{A}_M^t$ serve as the inputs for G to predict the OD demand matrix $\hat{\boldsymbol{M}}_{t+1}$ for the next time interval. Following this, the conditional information $\boldsymbol{L}_t$ and $\boldsymbol{A}_M^t$, the generated demand image $\hat{\boldsymbol{M}}_{t+1}$, the actual demand image $\boldsymbol{M}_{t+1}$ and the corresponding random linear interpolation $\tilde{\boldsymbol{M}}_{t+1}$ are fed into D. Finally, the output of D is modified using Wasserstein divergence to improve the overall performance of the model.

The initial GAN structure comprises stacked neural networks designed to fit a simple data distribution. Inspired by DCGAN [26], CNNs are utilized as the primary structure for both the generator and discriminator, performing feature extraction on OD demand images. Furthermore, ResNet [42] is incorporated to enhance the model performance, significantly improving the expression ca-

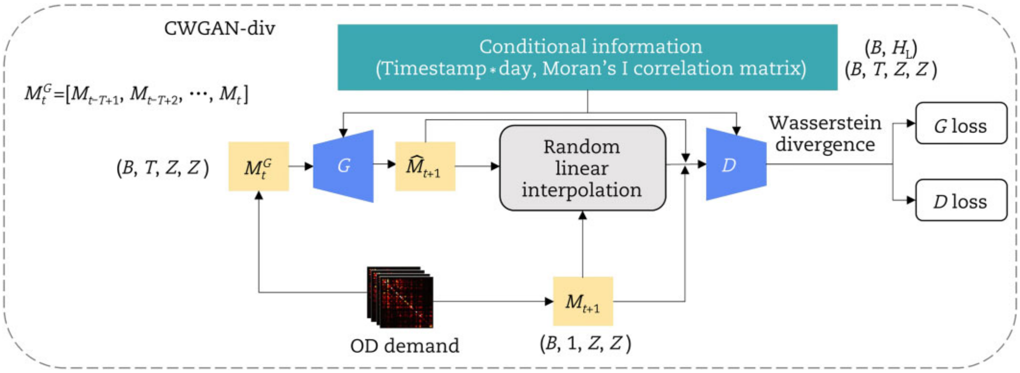

Fig. 3. Internal structure of CWGAN-div.

pacity of neural networks and preventing degradation. Fig. 4 displays the detailed components of the generator G and the discriminator D, and the specific structures of the residual block and fully connected (FC) block in G and D.

G is composed of three submodules, each responsible for calculating OD demand, temporal dependencies and spatial dependencies. The outputs from these submodules are then combined and fed into a shared layer for generating predictions. Likewise, D is made up of three submodules designed to handle the mixed OD demand and conditional data. The outputs from these submodules are calculated in an analogous manner.

To streamline the presentation of G and D implementation methods, we first delineate the computational operations of several fundamental layers and blocks. The BatchNorm layer is employed to expedite the training process by normalizing the output prior to its input into the subsequent layer. The BatchNorm layer's computational procedure can be defined as follows:

$$h_{i+1} = \frac{h_i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon_i}} \gamma_i + \beta_i \tag{10}$$

where μ_i and σ_i represent the mean and standard deviation of h_i, respectively. The parameters γ_i and β_i are learned by the neural networks, and ϵ_i is a constant. Furthermore, the Dropout layer is utilized to randomly deactivate neurons, thereby averting overfitting. Generally, the Dropout layer is incorporated following a convolutional or FC layer and can be represented as:

$$h_{i+1} = \Phi\left(w \cdot h_i + b\right) \cdot r \tag{11}$$

where Φ denotes the activation function, while w and b represent the weights and biases, respectively. The vector r consists of independent Bernoulli random variables, each having a probability p of being equal to one. Additionally, we employ two activation functions: tanh and leaky rectified linear unit (LeakyReLU) functions, which can be described as:

$$\mathrm{Tanh}(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{12}$$

$$\mathrm{LeakyReLU}(x) = \max\left(\alpha x, x\right) \tag{13}$$

The residual block's layered structure (refer to Fig. 4(c)) comprises two convolution layers, two BatchNorm layers, two activation functions and one dropout layer. The first convolution layer, first BatchNorm layer, LeakyReLU layer and dropout layer's calculations can be described as follows:

$$h_{conv1} = \sum_{u=1}^{U} h_i * f_1\left(u\right) + b_{conv1} \tag{14}$$

$$h_{B1} = \frac{h_{conv1} - \mu_{conv1}}{\sqrt{\sigma_{conv1}^2 + \epsilon_{B1}}} \gamma_{B1} + \beta_{B1} \tag{15}$$

$$h_{LReLU1} = \Phi_{LReLU}\left(h_{B1}\right) \tag{16}$$

$$h_{dropout} = h_{LReLU1} \cdot r \tag{17}$$

where h_i and h_{conv1} represent the input and output of the first convolution layer, respectively. The filter size is represented by U, while b_{conv1} indicates the bias. $f(u)$ refers to the u^{th} filter of the convolution layer, and '$*$' signifies the convolution operation. In Equation (15), h_{B1} signifies the output of the BatchNorm layer, while μ_{conv1} and σ_{conv1} denote the mean and standard deviation of h_{conv1}, respectively. ϵ_{B1} is a constant added for numerical stability. The BatchNorm layer's parameters are denoted by γ_{B1} and β_{B1}. In Equation (16), h_{LReLU1} is the output of the first activation layer, and Φ_{LReLU} indicates the nonlinear activation operation in the LeakyReLU layer. In Equation (17), $h_{dropout}$ is the Dropout layer's output. The subsequent convolution layer, BatchNorm layer and LeakyReLU layer are computed in the same manner as in Equations (14), (15) and (16), respectively.

During the convolution layer processing, the data's dimensionality might be altered. As a result, the shortcut connection cannot function as a direct mapping and necessitates processing via a convolution layer and a normalization layer. The calculations resemble Equations (14) and (15). Subsequently, the stacked layer output $h_{stacked}$ and the shortcut connection $h_{shortcut}$ are combined and processed through an activation function. This can be expressed as:

$$h_{out}^r = \Phi_{LReLU}\left(h_{shortcut} + h_{stacked}\right) \tag{18}$$

where h_{out}^r represents the residual block's output. The utilization of multiple convolution layers enables more effective capturing of spatiotemporal correlations in the OD demand data. Additionally, the residual mechanism averts the degradation problem and fosters convergence of CWGAN-div.

As the *Timestamp∗Day* variable is in the form of one-hot encoding, containing many zeros and exhibiting high sparsity, the structure is somewhat redundant and may affect the gradient updates during the process of CWGAN-div model training. Consequently, an FC block is employed to map *Timestamp∗Day* to a higher-dimensional space, enabling it to retain more valuable information (refer to Fig. 4(d)). The FC block comprises an FC layer, a BatchNorm layer and a LeakyReLU activation function layer, which can be computed as follows:

$$h_f = w_f \cdot h_i + b_f \tag{19}$$

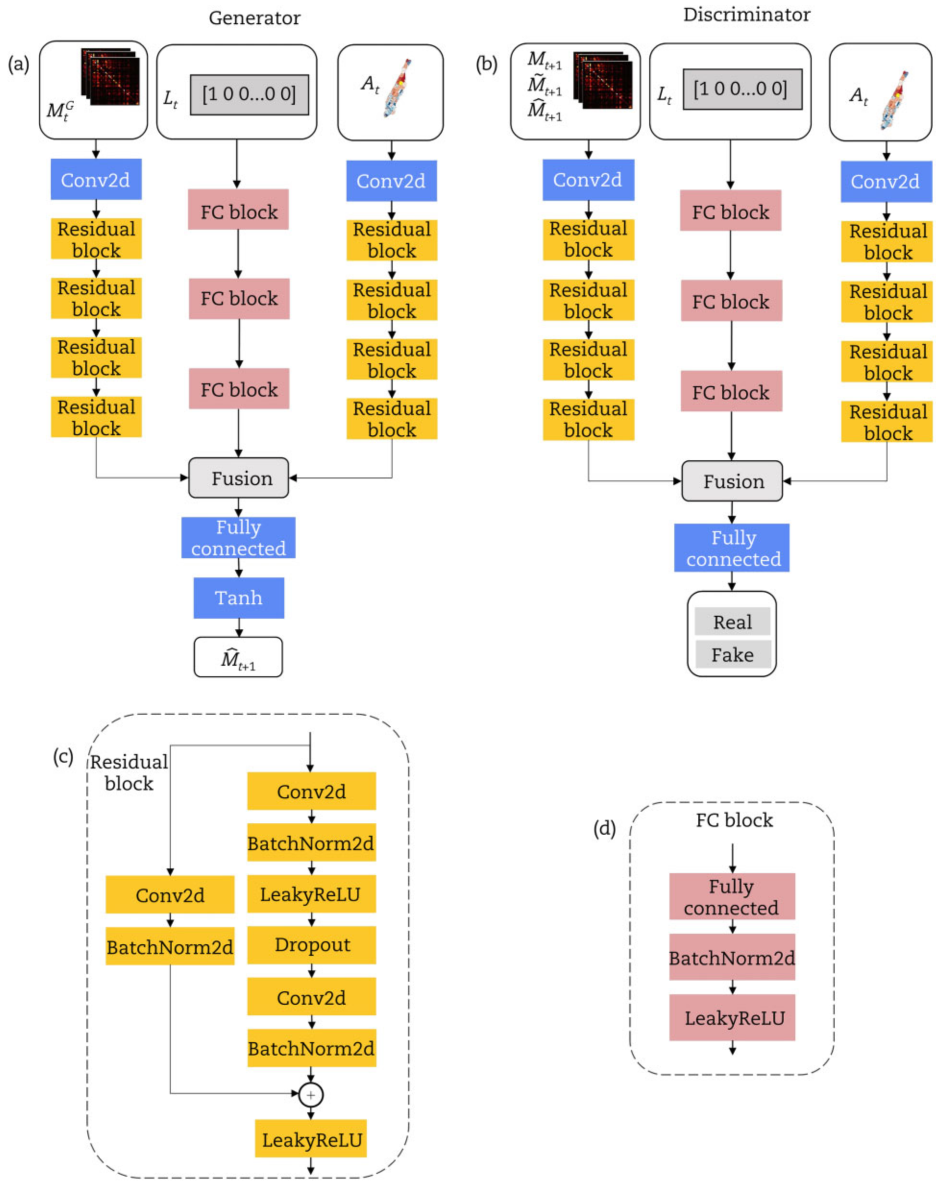

Fig. 4. Structures of (a) generator G, (b) discriminator D, (c) residual block and (d) FC block.

$$h_{out}^{FC} = \Phi_{LReLU}\left(\frac{h_f - \mu_f}{\sqrt{(\sigma_f)^2 + \epsilon_B}}\gamma_B + \beta_B\right) \tag{20}$$

where h_f and h_i represent the output and input of the FC layer, w_f and b_f denote the weights and bias terms. The output h_{out}^{FC} of the FC block can be obtained from Equation (20) by combining the above calculation method for the BatchNorm layer with the LeakyReLU activation function.

Having illustrated the inner workings of CWGAN-div and provided a brief description of the calculation processes for its fundamental components, the computation procedures of G and D are detailed further.

3.2. Computation procedures

The network structure of the generator G is similar to that of the discriminator D. The specific calculation process can be categorized into four parts as follows:

Part I. Processing OD demand data

The structure of this part comprises one convolution layer and four residual blocks. Referring to Equation (14) for the convolution operation, a convolution layer and an activation function form the input layer, which is formulated as

$$h_1^I = \Phi_{LReLU}\left(\sum_{u=1}^{U} h_0^I * f_1^I(u) + b_1^I\right) \tag{21}$$

where h_0^I represents the input of Part I. In generator G, h_0^I denotes the historical OD demand; in discriminator D, h_0^I signifies the mixed OD demand.

Following the convolution layer, the output is passed through a sequence of four residual blocks. The calculation process for the first residual block is

$$h_2^I = \Phi_{LReLU}\left(F_1^I\left(h_1^I\right) + R_1^I\left(h_1^I\right)\right) \tag{22}$$

where F_1^I denotes the convolution operation of the shortcut connection, and R_1^I signifies the residual function, in accordance with Equations (14) to (18). Upon completion of calculation through the four residual blocks, the output h_5^I can be obtained.

Part II. Processing temporal conditional information

This part of the network is employed in both G and D to process the *Timestamp∗Day* variable. The structure comprises three FC blocks with the *Timestamp∗Day* variable as input. Referring to Equation (20), the output h_1^{II} of the first FC block can be determined as

$$h_1^{II} = \Phi_{LReLU}\left(\frac{\left(w_1^{II} \cdot L_t + b_1^{II}\right) - \mu_1^{II}}{\sqrt{\left(\sigma_1^{II}\right)^2 + \epsilon_1^{II}}}\gamma_1^{II} + \beta_1^{II}\right) \tag{23}$$

Following the same three FC block procedures, we obtain h_3^{II}.

Part III. Processing spatial conditional information

This part of the network is employed in both G and D to process the Moran's I correlation matrix. Owing to the comparable dimensionality of the spatial correlation matrix and the OD demand matrix, the network structure is identical to that of Part I, comprising a convolution layer and four residual blocks. Referring to Equation (22), the computational procedure for the input layer is given as

$$h_1^{III} = \Phi_{LReLU}\left(\sum_{u=1}^{U} h_0^{III} * f_1^{III}(u) + b_1^{III}\right) \tag{24}$$

where h_0^{III} represents the input to Part III, i.e. the Moran's I correlation matrix $\mathbf{A}_M^t$.

Following this, the convolution layer output h_1^{III} is fed into the next four residual blocks, with the first residual block calculated as

$$h_2^{III} = \Phi_{LReLU}\left(F_1^{III}\left(h_1^{III}\right) + R_1^{III}\left(h_1^{III}\right)\right) \tag{25}$$

Similar to h_5^I in Part I, h_5^{III} is obtained after processing with four residual blocks.

Part IV. Fusion and output

This part of the network is utilized in both G and D to merge the outputs from the first three parts of the network, which are subsequently processed to produce the final output for generator G and discriminator D. The output of the generator represents the OD demand for the next interval, and the discriminator's output yields a score that represents the authenticity of the input information. The calculation process is as follows: the flattened outputs of the first three parts (i.e. h_5^I, h_3^{II} and h_5^{III}) are merged and subsequently passed into the FC layer for conversion into the target dimension output.

For generator G, a tanh activation function is employed after the FC layer to yield the predicted value $\hat{\mathbf{M}}_{t+1}$, calculated as

$$\hat{\mathbf{M}}_{t+1} = \mathrm{Re}\left(\Phi_{\mathrm{Tanh}}\left(w_G \cdot \left(\mathrm{Flat}\left(h_5^I\right); \mathrm{Flat}\left(h_3^{II}\right); \mathrm{Flat}\left(h_5^{III}\right)\right) + b_G\right)\right) \tag{26}$$

where Re signifies the reshaping operation converting the one-dimensional output of the FC layer into a two-dimensional OD matrix, Φ_{Tanh} represents the tanh activation function and Flat refers to the operation that flattens the data into one dimension.

For discriminator D, no activation function follows the FC layer, as determined by the properties of WGAN-div, and thus the output of discriminator h_D is computed as

$$h_D = w_D \cdot \left(\mathrm{Flat}\left(h_5^I\right); \mathrm{Flat}\left(h_3^{II}\right); \mathrm{Flat}\left(h_5^{III}\right)\right) + b_D \tag{27}$$

3.3. Training process of CWGAN-div

In the preceding subsection, the comprehensive computational methods employed for G and D are delineated. This subsection expounds upon the training process of CWGAN-div. Based on Equations (7) and (9), we can derive the objective function for CWGAN-div as follows:

$$\begin{aligned}\min\max V(G, D) = &\ E_{x\sim P_r}\left[\log\left(D\left(x|c\right)\right)\right] - E_{G(z)\sim P_g}\left[D\left(G\left(z|c\right)|c\right)\right] \\ &- kE_{\tilde{x}\sim P_u}\left[\| \nabla_{\tilde{x}} D\left(\tilde{x}|c\right) \|^p\right]\end{aligned} \tag{28}$$

Consequently, throughout the updating processes for G and D, the subsequent loss functions of G and D are minimized:

$$L\left(G\right) = -E_{G(z)\sim P_g}\left[D\left(G\left(z|c\right)|c\right)\right] \tag{29}$$

$$\begin{aligned}L\left(D\right) = &-E_{x\sim P_r}\left[D\left(x|c\right)\right] + E_{x\sim P_g}\left[D\left(x|c\right)\right] \\ &+ kE_{\tilde{x}\sim P_u}\left[\| \nabla_{\tilde{x}} D\left(\tilde{x}|c\right) \|^p\right]\end{aligned} \tag{30}$$

Generally, P_u utilizes the random linear interpolation of actual and generated samples. Initially, a set of actual samples, generated samples and random numbers conforming to a uniform distribution are randomly chosen, i.e. $x_r \sim P_r$, $x_g \sim P_g$, $\varepsilon \sim U(0, 1)$, and subsequently, the random linear interpolation is expressed as follows:

$$\tilde{x} = \varepsilon x_r + (1 - \varepsilon)\, x_g \tag{31}$$

Here is a brief summary of the computational procedure for CWGAN-div:

Step 1: Sample the historical OD demand (i.e. $\mathbf{M}_t^G$) and the corresponding demand in the subsequent time interval (i.e. $\mathbf{M}_{t+1}$) from the database, and obtain the associated *Timestamp∗Day* variable $\mathbf{L}_t$ and the Moran's I correlation matrix $\mathbf{A}_M^t$.

Step 2: Generate the predicted demand $\hat{\mathbf{M}}_{t+1}$ by inputting $\mathbf{M}_t^G$, $\mathbf{L}_t$ and $\mathbf{A}_M^t$ into G. Moreover, obtain $\tilde{\mathbf{M}}_{t+1}$ through interpolation according to Equation (31).

Step 3: Input $\hat{\mathbf{M}}_{t+1}$, $\mathbf{M}_{t+1}$, $\tilde{\mathbf{M}}_{t+1}$ and the corresponding spatiotemporal conditional information $\mathbf{L}_t$ and $\mathbf{A}_M^t$ into D, and then obtain $D(\mathbf{M}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t)$, $D(\hat{\mathbf{M}}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t)$ and $D(\tilde{\mathbf{M}}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t)$.

Step 4: Compute the Wasserstein distance and the loss function of D by

$$W = E\left[D\left(\mathbf{M}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t\right)\right] - E\left[D(\hat{\mathbf{M}}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t)\right] \tag{32}$$

$$\begin{aligned}L(D) = &-E\left[D\left(\mathbf{M}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t\right)\right] + E\left[D\left(\hat{\mathbf{M}}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t\right)\right] \\ &+ kE\left[\|\nabla D\left(\tilde{\mathbf{M}}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t\right)\|^p\right]\end{aligned} \tag{33}$$

Step 5: Compute the loss function of G by

$$L(G) = -E\left(D\left(\hat{\mathbf{M}}_{t+1}|\mathbf{L}_t;\ \mathbf{A}_M^t\right)\right) \tag{34}$$

Step 6: Compute the gradients of $L(D)$ and $L(G)$, and then optimize D and G using the Adam adaptive learning rate optimization algorithm.

Step 7: Save the optimal model. Then, predict the OD demand in the subsequent time interval using the well-trained generator.

4. Experiments and results

4.1. Data description

The data utilized in this study was derived from New York City ride-hailing order records, published by the New York City Taxi and Limousine Commission between 1 August and 21 December 2018. These records contain information regarding the geographical locations of passenger pick-ups and drop-offs, and corresponding timestamps. The study site in Manhattan was divided into 63 zones, as shown in Fig. 5. After data cleaning, the OD

Fig. 5. Study site with 63 traffic zones.

demand was calculated at 30-minute intervals, and the associated temporal and spatial conditional data was recorded. Subsequently, data from August to October was designated as the training dataset, data from November as the validation dataset and the remaining data as the test dataset. In accordance with the model's requirements, Max-Min normalization was employed to scale the data to the range [−1,1] before training the CWGAN-div.

4.2. Impact of spatiotemporal information on model performance

To examine the influence of temporal and spatial condition information on prediction results, four comparative experiments are performed in this subsection, as outlined below:

1) WGAN-div: The network structures and loss functions of WGAN-div are similar to those of CWGAN-div, but without conditional information.
2) TWGAN-div: The network structures and loss functions of TWGAN-div are similar to those of CWGAN-div, but without external spatial dependencies in the conditional information.
3) STWGAN-div: The network structures and loss functions of STWGAN-div are the same as those of CWGAN-div, but the spatial conditional information is distance matrices.
4) CWGAN-div: The whole model is described in Section 3.

Concerning the details of CWGAN-div, the number of channels in the convolution layer of the generator processing OD demand is 32, and the number of channels in the four residual blocks are 32, 64; 64, 128; 128, 256; 256, 512. The number of hidden nodes in the FC layer of the generator processing temporal information sequentially follows 128, 64 and 32. The number of channels in the convolution layer of the generator processing spatial information is 16, with the number of channels in the four residual blocks being 16, 32; 32, 64; 64, 128; 128, 256. The parameters in the discriminator dealing with the OD matrix and conditional information align with those in the generator. The model employs the Adam optimizer with a learning rate of 2e-4.

Table 1. Comparison of prediction results with different temporal and spatial information.

Model	MAPE/%	RMSE
WGAN-div	33.46	1.78
TWGAN-div	29.19	1.69
STWGAN-div	27.67	1.63
CWGAN-div	24.52	0.99

The number of backtracking cycles T is set to four for all methods. The mean absolute percentage error (MAPE) and root mean square error (RMSE) are employed to assess model performance:

$$MAPE = \frac{1}{Q}\sum_{i=1}^{Q}\left(\frac{1}{N}\sum_{i=1}^{N}\frac{\left|y_t^i - \hat{y}_t^i\right|}{y_t^i}\right) \tag{35}$$

$$RMSE = \sqrt{\frac{1}{Q}\sum_{i=1}^{Q}\left(\frac{1}{N}\sum_{i=1}^{N}\left(y_t^i - \hat{y}_t^i\right)^2\right)} \tag{36}$$

where Q is the number of prediction cycles, N is the number of OD pairs, $\hat{y}_t^i$ is the predicted OD value and y_t^i is the actual OD value.

The comparison results are presented in Table 1. The addition of temporal condition information increases the model prediction accuracy by 4.3%, indicating that the OD demand exhibits a noteworthy temporal correlation and incorporating temporal information effectively guides the model towards generating better prediction results. Furthermore, integrating the distance matrix also enhances the model prediction accuracy to a certain extent, suggesting that passengers' decisions to use ride-sourcing services are influenced by the distance between origin and destination zones. The proposed CWGAN-div, which employs the Moran's *I* correlation matrix as the spatial information input, achieves optimal prediction results. This is attributed to the Moran's *I* correlation matrix encompassing not only distance information but also demand correlations among zones.

4.3. Prediction performance of CWGAN-div

Taking the OD demand from zone 161 to zone 186 as an example, Fig. 6 illustrates the close correspondence between the predicted results of the CWGAN-div in the test dataset and the actual values. This indicates the CWGAN-div can accurately capture the variation in OD demand. To evaluate the prediction results in detail, we illustrate the network-wide OD demand by the heat maps of the OD matrices with 63×63 pixels for the evening peak of 1 December from 18:00 to 18:30 (see Fig. 7). The brighter the colour of the pixel block, the greater the ride-sourcing OD demand between two zones, and vice versa. Most of the zones appear darker in colour, corresponding to demand values of zero or small numbers, thus verifying the high sparsity of the OD matrix. There are 3969 OD pairs spanning 63 zones, indicating the challenge of dealing with numerous demand forecasts. Overall, the generated images predicted by CWGAN-div are in good agreement with the actual ones, indicating the superior prediction performance of the CWGAN-div.

Subsequently, special cases in the prediction results are analysed, namely 0–1 errors and 1–0 errors. 0–1 errors refer to OD pairs with an actual demand value of zero being predicted as non-zero (see Fig. 7(c)). Conversely, 1–0 errors refer to OD pairs with a non-zero actual demand value being predicted as zero (see Fig. 7(d)). A bright pixel represents a special error for that OD pair, while a black pixel indicates the absence of such an error. The relatively small number of bright pixels associated with 0–1 errors

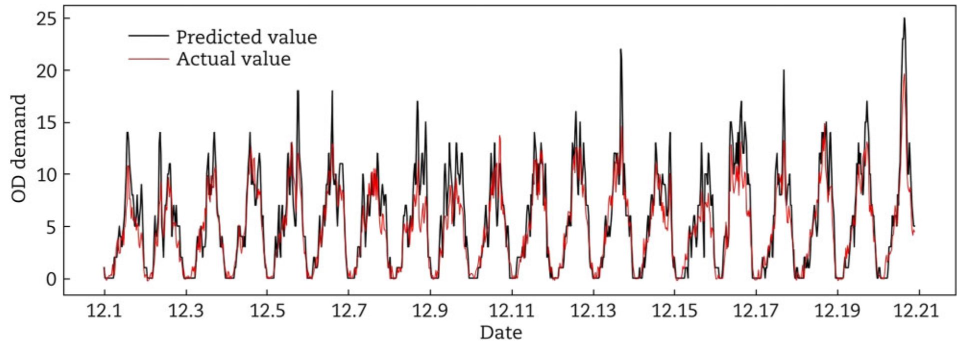

Fig. 6. Comparison of actual and predicted OD demand from zone 161 to zone 186.

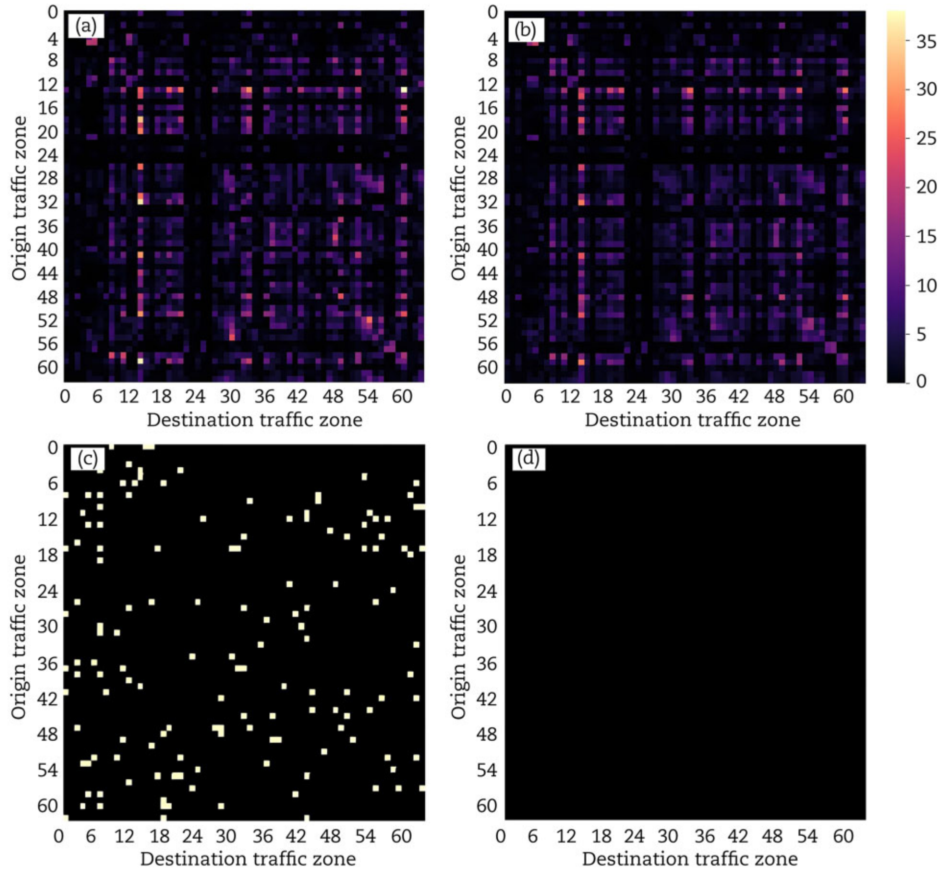

Fig. 7. Prediction results for the OD matrix from 18:00 to 18:30 on 1 December 2018: (a) actual OD matrix, (b) predicted OD matrix, (c) 0–1 prediction errors and (d) 1–0 prediction errors.

implies that their influence on real-world applications is likely to be insignificant. Furthermore, no 1–0 errors occur, indicating that CWGAN-div ensures the integrity of OD demand for the given time interval.

We further analyse the frequency of 0–1 errors and 1–0 errors in the test dataset. The empirical cumulative distribution function (CDF) curve in Fig. 8(a) shows that over 97% of the data does not exhibit 0–1 errors, and more than 99% of the 0–1 errors are below two. Similarly, Fig. 8(b) demonstrates that over 99.99% of the data does not present 1–0 errors, and even when 1–0 errors occur, all differences between predicted and actual values remain less than two. In practice, a negligible number of OD demands may be attributed to random factors, rendering accurate prediction of these demands insignificant. The above numerical results confirm the effectiveness of CWGAN-div in predicting network-wide ride-hailing OD demand.

4.4. Convergence of CWGAN-div

The aim of this subsection is to analyse the convergence of the CWGAN-div model during the training process, as illustrated in Fig. 9. Initially, the losses of the generator and discriminator fluctuate considerably, converging at approximately 400 epochs and ultimately stabilizing around zero. Conversely, the generator loss of the model without residual blocks persists in oscillating and struggles to converge. Although the discriminator loss can converge, its overall stability and convergence speed are inferior compared to the model incorporating residual blocks. This phe-

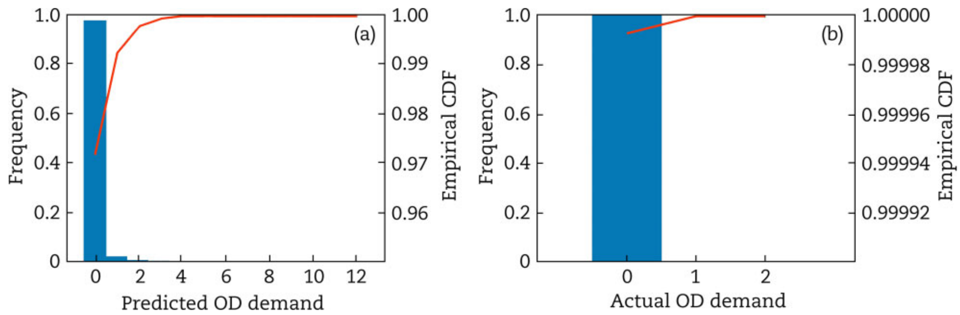

Fig. 8. Frequency of (a) 0–1 prediction errors and (b) 1–0 prediction errors in the test dataset.

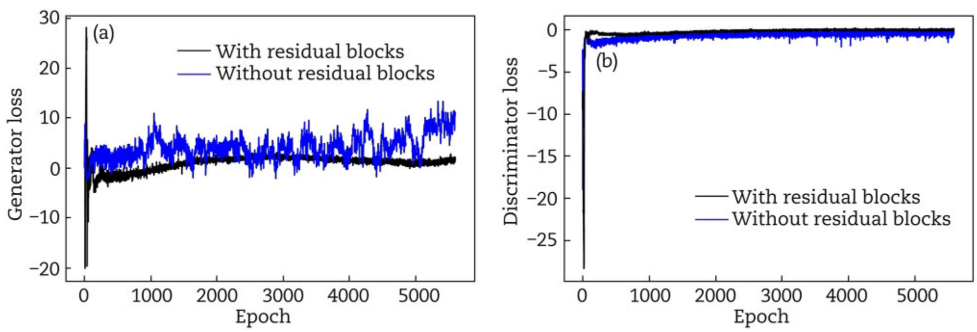

Fig. 9. Convergence of (a) the generator loss and (b) discriminator loss during the training process on the ride-hailing order data from the New York City Taxi & Limousine Commission.

nomenon can be ascribed to the fact that following multiple non-linear operations in the deep network, the gradient tends to vanish, adversely affecting the convergence of the generator and discriminator. These findings imply that the incorporation of residual blocks can enhance the learning capability of CWGAN-div and prevent network degradation.

The Wasserstein distance can be employed to measure the difference between the generated and actual distributions, as demonstrated by Fig. 10, which depicts the trend of the Wasserstein distance for CWGAN-div during the initial 800 training epochs. Specifically, the Wasserstein distance experiences significant fluctuations between 10 and 30 training epochs before gradually decreasing and converging to zero. This occurs because, in the early stages of training, the generator can generate only random OD demand matrices. As the number of training epochs increases, the generator's generative capability is effectively enhanced. Beyond the curve's peak, the generated OD matrix increasingly approximates the actual OD matrix. After 400 training epochs, the generated OD image forms more clearly, and the Wasserstein distance stabilizes near zero.

4.5. Model comparison

To thoroughly assess the effectiveness of CWGAN-div, this section conducts comparative experiments from three distinct perspectives. Firstly, the historical average (HA) method and CNN are chosen, with the former being a widely used statistical prediction approach and the latter a classic deep learning model. Secondly, a comparison is made with the combined deep learning model, CNN-LSTM [19]. Lastly, comparisons are drawn with similar GAN models, namely the initial GAN, CGAN and WGAN-div. The configurations of these six counterparts are as follows:

1) HA: The average historical demand is calculated as the OD demand forecast for the next interval.
2) CNN: The model structure comprises five convolution layers (with channel numbers of 32, 64, 128, 256, 512), five max-pooling layers and one FC layer.
3) CNN-LSTM: Five convolution layers (32, 64, 128, 256, 512) and five max-pooling layers are utilized for the CNN part, while three LSTM layers with 2048 neurons are employed for the LSTM part.
4) Initial-GAN: The network structures and inputs of initial GAN are similar to those of CWGAN-div, but without conditional information and residual blocks. Owing to the basic principle of initial GAN, the objective function is based on KL divergence.
5) CGAN: The network structures and loss functions of CGAN are similar to those of initial GAN, but the inputs include spatiotemporal conditional information.
6) WGAN-div: The network structures and loss functions of WGAN-div are similar to those of CWGAN-div, but without the spatiotemporal conditional information.

Table 2 presents the experimental results of the six baseline models and CWGAN-div. It is evident that CWGAN-div surpasses the others, followed by CNN-LSTM, CNN, HA and similar GANs. Specifically, the HA method offers convenient computation but struggles to cope with real-time fluctuating traffic demand. CNN, a deep learning model widely employed in image processing, significantly enhances the prediction performance; however, the pure CNN model is less effective in mining the traffic data's spatiotemporal characteristics and exhibits a black-box and poorly interpretable nature. The CNN-LSTM model combines the advantages of LSTM and CNN, improving the prediction performance by

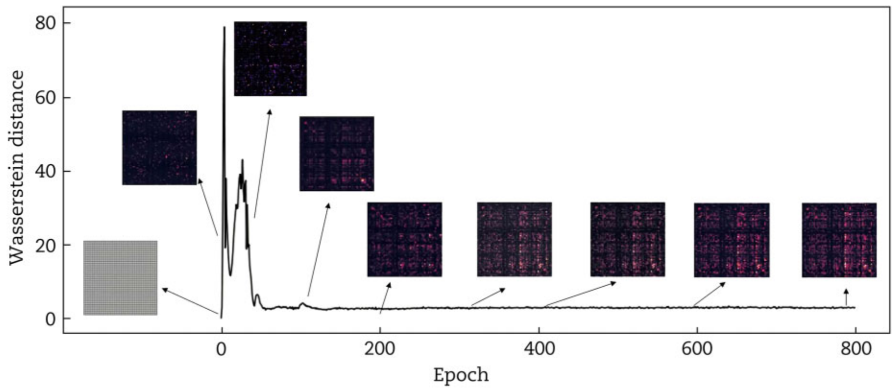

Fig. 10. Training process of CWGAN-div on the ride-hailing order data from the New York City Taxi & Limousine Commission.

capturing both temporal and spatial dependencies, yet its error is still 3.01% higher than CWGAN-div. Among the similar GAN models, the initial GAN performs the worst, mainly due to not utilizing an objective function based on the Wasserstein distance metric, which makes the model prone to training instability and vanishing gradients. However, the prediction accuracy of CGAN improves compared to the initial GAN because the external spatiotemporal data guides the model to generate better results. CWGAN-div combines the benefits of CGAN and WGAN-div, incorporating not only interpretable conditional information but also an advanced network structure with an improved objective function. The effective implementation of CWGAN-div highlights that utilizing Wasserstein divergence, external spatiotemporal dependencies and residual blocks can boost the stability during model training while notably increasing the OD demand prediction accuracy.

5. Conclusions

In this paper, we propose a CWGAN-div model to predict ride-hailing OD demand matrices with large-scale, high-dimensional and sparse characteristics. The proposed CWGAN-div model combines the strengths of both CGAN and WGAN-div, effectively capturing the internal spatiotemporal correlations of OD demand while ensuring training stability. Moreover, the model leverages external spatiotemporal dependencies as conditional information to guide the generation of highly accurate predictions. Utilizing ride-hailing order data from Manhattan, New York City, we assess the influence of spatiotemporal information on model performance through numerical experiments. Our results indicate that the *Timestamp∗Day* variable and the Moran's I correlation matrix enhance prediction accuracy. Additionally, we conduct an in-depth analysis of the generated spatiotemporal images by examining 0–1 errors and 1–0 errors. Numerical results demonstrate that the OD matrices predicted by CWGAN-div are in good agreement with the actual ones. Ablation experiments show that integrating residual blocks improves the training stability and expedites convergence. Compared to other prediction approaches (HA, CNN, CNN-LSTM, GAN, CGAN and WGAN-div), the proposed CWGAN-div model achieves superior prediction performance. In conclusion, the CWGAN-div model has considerable potential for network-wide ride-hailing OD demand prediction. Future research may explore diverse datasets to evaluate the robustness and generalization of CWGAN-div, as well as investigating appropriate dimensionality reduction techniques to deal with data redundancy by inputting low-dimensional and dense representations of spatiotemporal images.

Table 2. Comparison of prediction results with the six baseline models.

Model	MAPE/%	RMSE
HA	32.74	1.89
CNN	28.19	1.53
LSTM-CNN	27.53	1.45
GAN	56.85	3.89
CGAN	37.92	2.57
WGAN-div	33.46	1.78
CWGAN-div	24.52	0.99

Acknowledgements

This work has been supported by the National Natural Science Foundation of China (Grant No. 72371251), the National Science Foundation for Distinguished Young Scholars of Hunan Province (Grant No. 2024JJ2080), the Excellent Youth Foundation of Hunan Education Department (Grant No. 21B0015) and the State Key Laboratory of Rail Traffic Control and Safety of Beijing Jiaotong University, China (Grant No. RCS2022K004).

Conflict of interest statement

The authors declare that they have no conflicts of interest.

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Received: April 23, 2023. **Revised:** May 12, 2023. **Accepted:** May 13, 2023